



MEDICAL AI SYSTEMS: MRI ANALYSIS

Role of Computer Vision, Physics of Image Formation & Clinical Relevance

Presented by
P.YASWANTH
192524167

Overview: AI in Modern Radiology

Why AI for MRI?

- Radiologists review thousands of images daily.
- AI acts as a 'Second Pair of Eyes'.
- Faster triage: Flags urgent cases (e.g., Stroke, Tumors) immediately.

AI Automated Detection (Computer Vision)

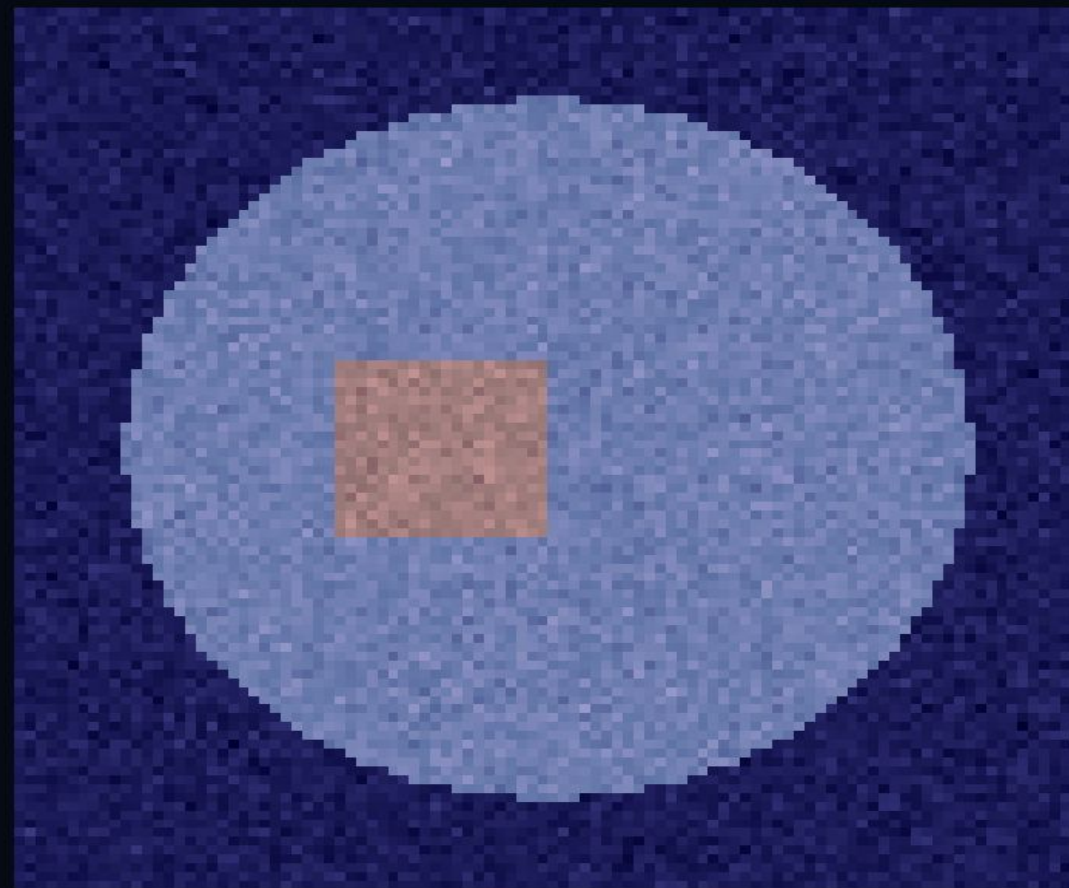


IMAGE FORMATION: How MRI is Built

1. Nuclear Magnetic Resonance:

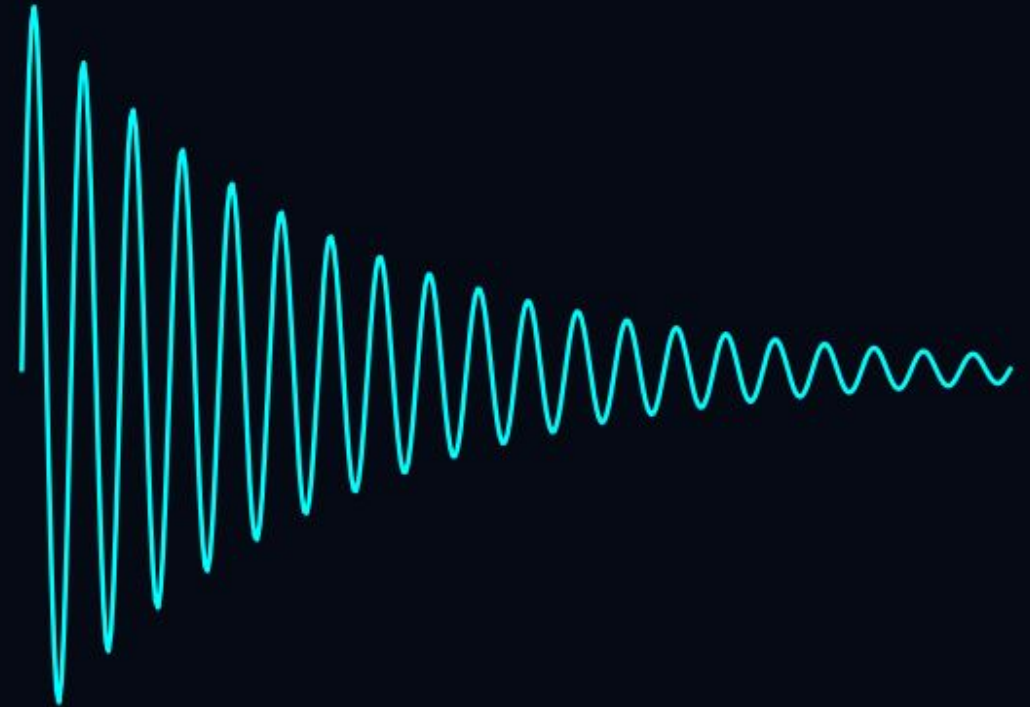
Protons in water align with a strong magnetic field.

2. RF Pulse: Energy is sent to flip the protons.

3. Relaxation & Signal: As protons flip back, they emit a signal (FID).

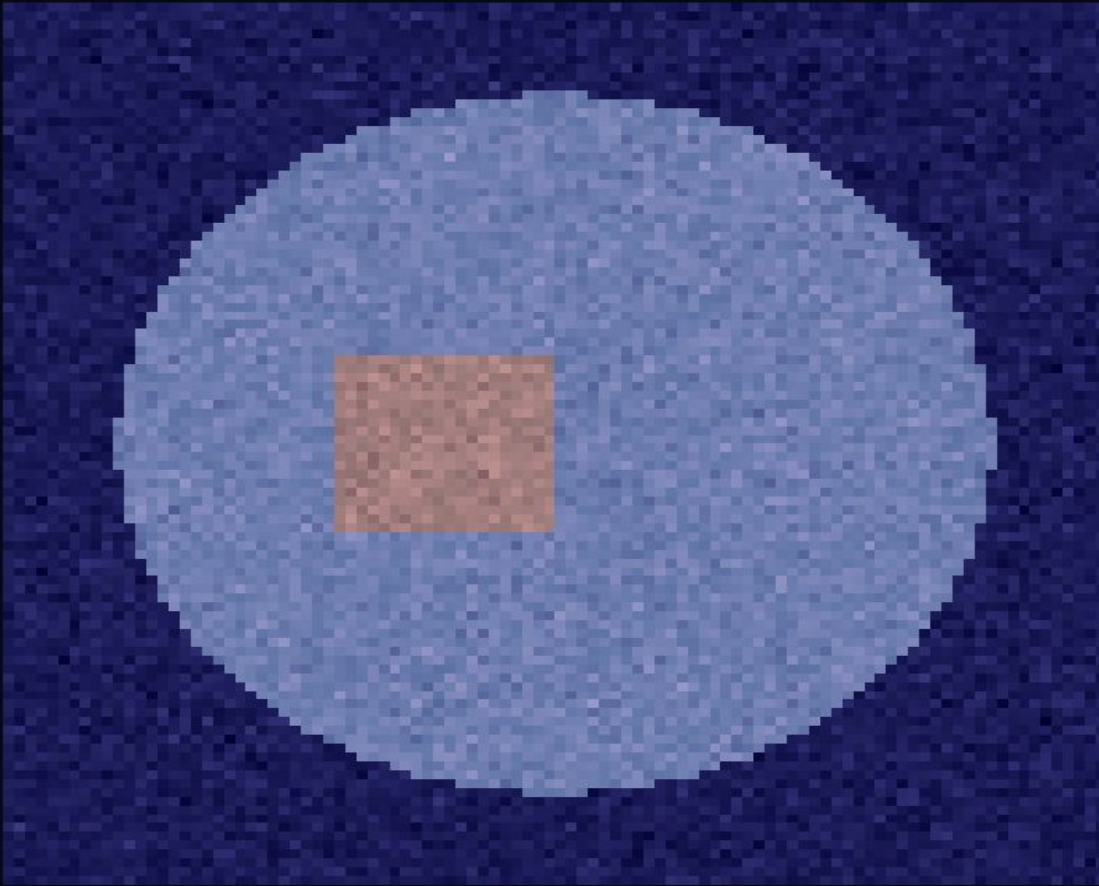
4. K-Space to Image: Signals are stored in K-Space (Fourier Transform) to create the image pixels.

MRI Signal Acquisition (NMR Physics)



THE ROLE OF COMPUTER VISION

AI Automated Detection (Computer Vision)



Computer Vision 'sees' patterns radiologists might miss.

- Feature Extraction:

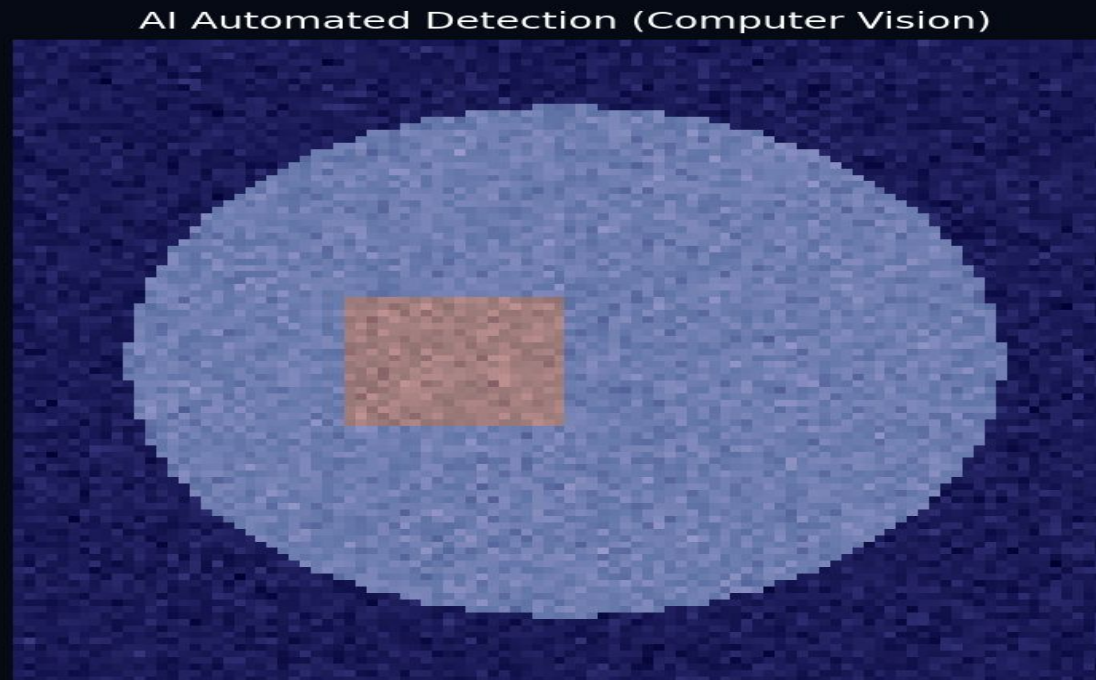
Identifies edges, textures, and tissue density.

- Texture Analysis:

Determines if a tumor is malignant (dangerous) or benign based on pixel consistency.

Segmentation: Mapping the Anatomy

- Semantic Segmentation: Every pixel is labeled (e.g., Gray matter vs White matter).
- Object Detection: Bounding boxes around suspicious lesions.
- Result: Precise 3D measurement of tumor volume for treatment planning.



How the AI 'Decides': The Workflow

Pre-processing

Cleaning noise from
MRI physics.

CNN Model

Artificial Neural
Network processes
pixels.

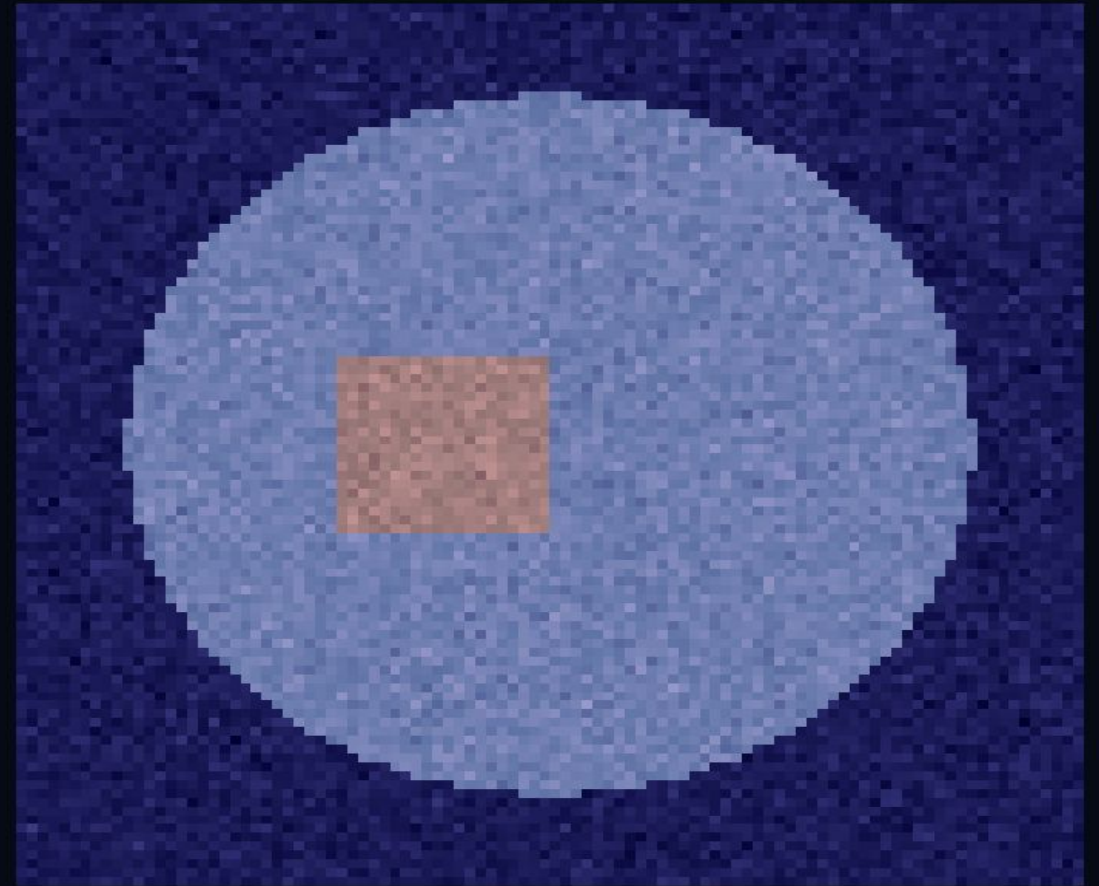
Diagnosis

Probabilistic scoring of
disease.

APPLICATION RELEVANCE: Impact on Health

- Neurology: Detecting early Alzheimer's by measuring brain shrinkage.
- Oncology: Monitoring if a tumor shrinks after chemotherapy.
- Cardiology: Analyzing real-time MRI videos of heart valves.
- Cost-Efficiency: Reduces the need for repeat scans.

AI Automated Detection (Computer Vision)



In Summary: AI transforms MRI from a static picture into dynamic data.

Physics creates the image; Computer Vision understands it.

Thank You!



Kuthambakkam, Tamil Nadu, India 🇮🇳

Cse & It Department, Kuthambakkam, Tamil Nadu 602105, India

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